

1. **Dial-up networking**
 - A. Is a full-featured communications program for accessing mainframes?
 - B. Supports 12 terminal emulations
 - C. Provides a communications connection to an internet service provider
 - D. Enable file transfers between mainframe using 14 different file-transfer protocols
2. **What is the name of the network topology in which there are bi-directional links between each possible node?**
 - A. Ring
 - B. Tree
 - C. Star
 - D. Mesh
3. **An example of CSMA/CD standard is:**
 - A. Ethernet
 - B. FDDI
 - C. Token ring
 - D. All of the above
4. **What is a usual number of bits transmitted simultaneously in parallel data transmission used by microcomputer?**
 - A. 6
 - B. 8
 - C. 9
 - D. 7
5. **When a host knows its physical address but not its IP address, it can use**
 - A. ICMP
 - B. ARP
 - C. IGMP
 - D. RARP
6. **Hardwired implementation is:**
 - A. CISC
 - B. RISC
 - C. Both of the above
 - D. None of the above
7. **ISA stands for:**
 - A. Industry Standard Architecture
 - B. Intel Standard Architecture
 - C. Interconnect Standard Architecture
 - D. None of the above
8. **The memory allocation scheme subject to "external" fragmentation is**
 - A. Segmentation
 - B. Pure demand paging
 - C. Swapping
 - D. Multiple contiguous-fixed partitions
9. **The problem of thrashing is affected significantly by:**
 - A. Program structure
 - B. Primary storage-size
 - C. Program size
 - D. All of the above
10. **Lexical analysis in Compiler.**
 - A. Analyze the links between the valid statements
 - B. Analyze the validity of tokens
 - C. Analyze the code for optimization
 - D. Loads the code in memory
11. **Which of the following is an example of government-to-citizen (G2C) E-Commerce?**
 - A. E-procurement
 - B. E-auctions
 - C. Electronic benefits transfer (EBT)
 - D. E-purchasing
12. **EPS stands for:**
 - A. Electronic Payable System
 - B. Electronic Payment System
 - C. Electronic-Marketing Payment System
 - D. All of the above

13. What does it mean by payment processor in E-commerce?
 - A. Credit card company
 - B. Company appointed by merchant to handle credit card transactions for merchant banks
 - C. A company which pays the transaction on behalf of merchant bank
 - D. Merchant
14. The symbol ":("represent the feelings and views over internet chatting as given below:
 - A. Smiling
 - B. Winking
 - C. Disappointing
 - D. Raising brown
15. Which of the following is NOT the valid symmetric algorithm for cryptography?
 - A. DES
 - B. IDEA
 - C. 3DES
 - D. Diffie- Hellman
16. Uniform Resource Locator (URL) are:
 - A. The same as search engine
 - B. Used to find specific documents and resource on the WWW.
 - C. Never more than 11-character long
 - D. All of the above
17. Which of the following are names for the internet?
 - A. The net
 - B. Information super highway
 - C. Cyberspace
 - D. All of the above
18. Which protocol is used to transfer information between WWW server and a WWW browser?
 - A. FTP
 - B. IP
 - C. TCP
 - D. HTTP
19. The internet is able to send information back and forth to different types of computers because of what software?
 - A. Application software
 - B. Packaged software
 - C. Operating system software
 - D. All of the above
20. A website that is restricted to the members of an organization i.e. within a collage, is called:
 - A. An intranet
 - B. A spared
 - C. An extranet
 - D. All of the above
21. Which of the following is a search engine?
 - A. Net scape
 - B. Alta Vista
 - C. Java
 - D. Yahoo.com
22. The internet was developed in the.....
 - A. Early 1970s
 - B. Late 1980s
 - C. Late 1960s
 - D. Early 1990s
23. World wide web pages can be described as multimedia pages, this means that a.....
 - A. The pages may contain
 - B. Text, pictures, sound
 - C. Text and pictures only
 - D. Video clips sound, text, pictures
24. The speed for the data pathway or channel is called
 - A. Channel speed
 - B. Disk access time
 - C. Baud rate
 - D. Data transfer rate
25. HTML stands for:
 - A. Hyper Test Markup Language
 - B. Hyper Text Markup Language
 - C. Hyper Test Media Language
 - D. None of the above

26. IT policy advisory body of Government of Nepal is:
- Ministry of Science and Government of Nepal is:
 - Ministry of Information and Communication
 - High Level Commission for Information Technology
 - None of the above
27. What is the name of block diagram that outlines, step, how a computer program will work?
- Program flow chart
 - Structured program
 - Program layout
 - Logical planning
28. What is the primary requisite of a good computer programmer?
- Mathematical background
 - Artistic mind
 - Scientific knowledge
 - Logical mind
29. National Computer Centre was established in Nepal in the year:
- 2028 BS
 - 2032 BS
 - 2031 BS
 - None of the above
30. According to cyber law of Nepal any one of the following meaning does not belongs to the definition of "Computer Samagree".
- Any information
 - Any hardware
 - Any software
 - Any human ware
31. The registrar, in addition to the powers mentioned in the act or the rules made there under, shall not have one of the following powers similar to court.
- To make necessary inquiry after administering oath
 - To compel search of any document
 - To requisition copy or record of any document from any office
 - To fine extending from Rs. 1000 to 50000 for making any unauthorized publication
32. The first copy right Act of Nepal was promulgated in the year:
- 1967
 - 2002
 - 1997
 - 2008
33. The function of the IT park development committee of Nepal is:
- To construct sub-IT part at any city of Nepal
 - To co-ordinate the sub-IT part
 - To implement info-cities
 - All of the above
34. Government integrated data center of Nepal is mandated to:
- Store and manage all the data and services of government of Nepal
 - Host the E-government services
 - Work as government ISP
 - All of the above
35. An application used in an organization which runs on a managed server external to the company is best known as:
- Packaged software
 - Bespoke Software
 - An application Service provider solution
 - Object-oriented software

36. Firewalls are used to protect against ____
- Unauthorized Attacks
 - Data Driven Attacks
 - External resources
 - Fire Attacks
37. The 4-bits binary number 0111 represents.
- 15
 - 7
 - 7
 - 1
38. The ASCII code of '0' (Zero) is:
- 48D
 - 0011 1000
 - 32H
 - 42H
39. A gigabyte represents.
- 1 billion bytes
 - 230 bytes
 - 1000 kilo bytes
 - 1024 bytes
40. Personal computers use number of chips mounted on a main circuit board. What is the common name for such boards?
- Daughter board
 - Father Board
 - Motherboard
 - Bread board
41. What is the control unit's function in the CPU?
- To store program instruction
 - To perform logic operations
 - To transfer data to primary storage
 - To decode program instruction
42. The storage required for an image such as an X-ray is approximately.
- A few bytes
 - A few gigabytes
 - A few hundred bytes
 - A few megabyte
43. In analog computer.
- Input is first converted to digital form
 - Output is displayed in digital form
 - Input is never converted to digital form
 - All of the above
44. The data recording format in most of the modern magnetic tape is:
- 7-bit ASCII
 - 8-bit ASCII
 - 7-bit EBCDIC
 - 8-bit EBCDIC
45. Computer follows a simple principle called GIGO which means.
- Garbage input good output
 - Great instruction great output
 - Garbage in Garbage Out
 - Good input good output
46. A data structure where elements can be added or removed at either end but not in the middle.
- Linked lists
 - Queues
 - Stacks
 - Dequeue
47. Which of the following sorting algorithm is of divide and conquer type?
- Bubble sort
 - Quick sort
 - Insertion sort
 - None of the above
48. A variable P is called pointer if:
- P contains the address of an element in DATA
 - P points to the address of first elements in DATA
 - P can store only memory addresses
 - P contain the DATA and the address of DATA
49. Binary search algorithm cannot be applied to:
- Sorted linked list
 - Sorted linear array
 - Sorted binary trees
 - Pointer array

50. A data structure where element can be added or removed at either end but not in the middle.
A. Linked lists B. Queues
C. Stacks D. Dequeue
51. Which one of the following is used to represent the stack empty condition?
A. $\text{Top} = \text{Max} - 1$ B. $\text{Top} = \text{Top} + 1$
C. $\text{Top} = 0$ D. $\text{Top} = -1$
52. The situation when in a linked list START-NULL is:
A. Underflow B. Houseful
C. Overflow D. Saturated
53. The time factor when determining the efficiency of algorithm is measured by:
A. Counting microsecond's
B. Counting the number of statements
C. Counting the number of key operations
D. Counting the kilobytes of algorithm
54. A physical DFD.
A. Can show only stored material
B. Can show the flow of material
C. Has no means of showing material flow
D. Does not concern itself with material flow
55. CASE tools are:
A. A set of rules to be used during system analysis and design
B. Program, packages used during system analysis and design
C. A set of tools used by analysis
D. Needed for use case development
56. The context diagram treats entire system as:
A. A major process
B. A double process
C. A single process
D. None of the above
57. A "middle-tier" software and hardware combinations that lies between the web server and the corporate network and system best defines:
A. Web server
B. Intranet server
C. Application server
D. Tiered server
58. In order to obtain financing for the analysis phase, the systems analyst must.
A. Justify the expense of upgrading
B. Consider abandoning the project
C. Prepare a preliminary investigation report
D. Train user on the new system
59. A candidate key that has been selected as the unique, identifying characteristics for an entity type best described a (n):
A. Data marker B. Associative
C. Identifier D. Primary attribute
60. They are not divided into subparts are called...
A. Single-valued attributes
B. Composite attributes
C. Simple attributes
D. Multi-value attributes
61. Who is responsible for developing joint application design (JAD)?
A. Composite Associates
B. Software developer
C. IT experts
D. Different domain experts
62. The first step in the systems development life cycle is:
A. Analysis
B. Design
C. Problem/opportunity identification
D. Development and documentation

63. In this phase of the systems life cycle, new or alternative information systems are designed.
 A. Preliminary investigation
 B. Systems design
 C. Systems analysis
 D. Systems development
64. The process related to process control, file management, device management, information about system and communication that is requested by any higher-level language can be performed by _____.
 A. Editors B. System call
 C. Compilers D. Caching
65. In which year the first version of Linux was released?
 A. 1991 B. 1997
 C. 1995 D. 1999
66. What is a shell?
 A. It is a hardware component
 B. It is a part in compiler
 C. It is a command interpreter
 D. It is a tool in CPU scheduling
67. Which scheduler controls the degree of multiprogramming?
 A. Short term scheduler
 B. Middle term scheduler
 C. Long term scheduler
 D. None of the above
68. Which of the following file name extension suggests that the file is Backup copy of another file?
 A. TXT B. BAS
 C. COM D. BAK
69. Which is not an Operating System?
 A. Windows 95 B. Windows 3.1
 C. MS-DOS D. Oracle
70. Which algorithm avoids deadlock by not granting a request if that request will put the system in an unsafe state?
 A. Ostrich algorithm
 B. Dijkstra's algorithm
 C. Banker's algorithm
 D. None of the above
71. The Banker's algorithm is used.
 A. To rectify a deadlocks state
 B. To detect deadlock in operating system
 C. To prevent deadlock in operating systems
 D. None of the above
72. Virtual memory is....
 A. An extremely larger main memory
 B. An extremely large secondary memory
 C. An illusion of extremely large main memory
 D. A type of memory used in super computer
73. Which of the following will determine your choice of systems software for your computer?
 A. Is the applications software you want to use compatible with it?
 B. Is it expensive?
 C. Is it compatible with your hardware?
 D. Both A and C
74. You are running a query against a relational database. Referring to the scenario above, what clause or command, do you use in the query to help avoid a costly table scan?
 A. GROUP BY clause
 B. HAVING clause
 C. INDEX command
 D. WHERE clause
75. We say that a relation schema R is in first normal form (NF), if:
 A. The columns of all attributes of R are atomic
 B. The columns of all attributes of R are non-atomic
 C. The columns of single attributes of R are atomic
 D. Both A and B

76. What is a 'tuple'?

- A. A row or record in a database to record
- B. An attribute attached to a record
- C. Another name for the key linking different tables in a database
- D. A row or record in a database table with similar attributes

77. % is used to search information on the basis of pattern matching like:

- A. Sub-string-matching
- B. Character matching
- C. Both (A) & (B)
- D. None of the above

78. Functions include:

- A. COUNT, MIN OR
- B. AVG, IF, COUNT
- C. MAX, SUM, AND
- D. COUNT, SUM, AVG

79. What can a mandatory one to one relationship indicate?

- A. More entities are needed
- B. More attributes are needed
- C. The model should be demoralized
- D. The tables are not properly indexed

80. A relation that contains no multi-valued attributes, and has non-key attributes dependent on the primary key, but contains transitive dependencies is in which normal form?

- A. First NF
- B. Third NF
- C. Second NF
- D. Fourth NF

81. The data dictionary is:

- A. A utility program
- B. A list of database personnel
- C. Different from an index procedure
- D. An application generator

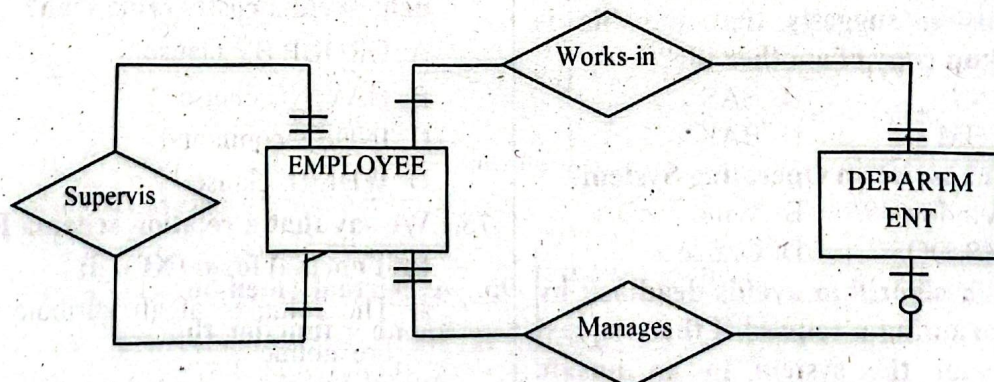
82. The _____ rule specifies that an entity can be a member of only one subtype at a time.

- A. Removal
- B. Disjoint
- C. Exclusion
- D. Inclusion

83. Which database subsystem compiles and executes DDL and DML statements?

- A. The Transaction Manager
- B. The storage manager
- C. The query processor
- D. None of the above

84. For the following diagram, which answer is true?



- A. Each employee can supervise 1 to many employees
- B. Each employee can manage many departments
- C. Each employee works in one department
- D. Both (A) & (C)

85. What is embedded SQL.
- A subset of SQL for Macintosh system
 - An SQL compiler built by Microsoft
 - SQL statements used in a program written in some other programming language, usually prefixed by some symbols recognizable by preprocessor
 - SQL written for embedded system, used in a program written in some other programming language
86. Which of the below undoes all the updates performed by the SQL statements in the transaction?
- Commit work
 - Rollback work
 - Both (A) & (C)
 - None of the above
87. An organized collection of logically related data is known as:
- Data
 - Database
 - Meta data
 - Data versus information
88. You are written a database application to run on your DBMS. You do not want your users to be able to view the underlying table structures. At the same time, you want to allow certain update operations.
- Referring to the above scenario, what structure will you deploy?
- Summary table
 - Dynamic procedure
 - Table filter
 - View
89. Which of the following languages is often translated to pseudo code?
- Assembly
 - PL/I
 - FORTTRAN
 - PASCAL
90. _____ is invoked to create an object.
- A method with a return type
 - A constructor
 - The main method
 - A method with the void return type
91. A computer-generated output that lets programmer follow the execution of their programs line by line is a.
- Core dump
 - Detail listing
 - Trace routine
 - Source listing
92. In a complete binary tree on n nodes, how far are the most distant two nodes? Assume that each edge in the path counts as 1. Assume $\log(n)$ is log base 2.
- About $\log(n)$
 - About $3 \log(n)$
 - About $2 \log_2(n)$
 - About $4 \log(n)$
93. When did the level programming language PASCAL become an ANSI standard language?
- 1983
 - 1985
 - 1980
 - 1987
94. A special purpose language developed for computer assisted instruction (CAI) is known as:
- LISP
 - PILOT
 - APL
 - SNOBOL
95. A virtual member function is a member function that can...
- Be overridden by a sub class
 - Be derived from another class
 - Move to any class
 - None of the above

96. Distance vector routing protocols
- A. Are good in case the network needs to be scalable
 - B. Has a constraint of "count to infinite" problem?
 - C. Overcomes the problem of "count to infinity" problem
 - D. Has no relation at all with "count to infinity" problem?
97. The dialogue techniques for terminal use do not include
- A. Question and answer
 - B. Form fillings
 - C. Open-ended questions
 - D. Menu display

98. UDP is better than TCP when.
- A. Error in transmission is likely to be higher
 - B. There should not be breakage in transmission
 - C. Loss in transmission is not considered critical
 - D. All the above
99. CRC method helps to
- A. Send the data without any error
 - B. Receive the data without any errors
 - C. Corrects the error during the transmission
 - D. Detect the error in digital network
100. Fiber-optic cables operate at frequencies near
- A. 20 MHz
 - B. 2G MHz
 - C. 200 MHz
 - D. 800 THz

ANSWER SHEET

1.C	2.D	3.A	4.C	5.D	6.B	7.A	8.A	9.A	10.B
11.B	12.B	13.B	14.C	15.D	16.B	17.B	18.D	19.D	20.A
21.B	22.A	23.D	24.C	25.B	26.B	27.A	28.D	29.A	30.D
31.D	32.B	33.D	34.D	35.C	36.A	37.B	38.A	39.A	40.C
41.D	42.D	43.C	44.D	45.C	46.D	47.B	48.A	49.A	50.D
51.D	52.A	53.C	54.B	55.B	56.C	57.C	58.C	59.C	60.C
61.D	62.C	63.B	64.B	65.A	66.C	67.C	68.D	69.D	70.C
71.C	72.C	73.D	74.C	75.A	76.A	77.C	78.D	79.B	80.C
81.C	82.C	83.C	84.D	85.C	86.C	87.B	88.D	89.D	90.B
91.C	92.C	93.A	94.B	95.A	96.B	97.B	98.D	99.D	100.D